CORRELATIONS BETWEEN PRIMITIVE ROOT STATUSES OF CONSECUTIVE PRIMES

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Abstract. Call a prime p an *Artin prime for base* 10 if 10 is a primitive root modulo p. We study, for the first time, the dependence between the Artin statuses of *consecutive* primes. Computing Artin status for all 50,847,531 primes up to 10^9, we find that consecutive primes are significantly *anticorrelated*: $P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) = 0.36514$ versus $P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n) = 0.37928$, a deficit $\delta = -0.01414$ ($z \approx -101$ over 50,847,529 consecutive pairs). The dependence has striking gap structure. We prove an elementary but, to our knowledge, unrecorded exclusion law: if p and $p + g$ are both prime with $g \equiv 20 \pmod{40}$, then 10 is a quadratic residue modulo exactly one of them, so they can never both be Artin primes for base 10; empirically, among 2,195,882 consecutive pairs with gap 20 or 60 there is indeed not a single doubly-Artin pair. Conversely, gaps $g \equiv 0 \pmod{40}$ force equal quadratic-residue status and exhibit strong positive dependence ($P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) = 0.899$ at $g = 40$). We then decompose the global anticorrelation into arithmetically forced channels: conditioning on the joint residues of (p_n, p_{n+1}) modulo 120 removes 93% of the effect, and conditioning modulo 840 removes 97%, with the residual dependence statistically compatible with the unabsorbed small-modulus channels ($\chi^2/\mathrm{df} = 1.14$ on 633 cells). That the decomposition closes so cleanly is itself a sharp consistency test, at 50 million pairs, of the Hardy–Littlewood-based explanation of the consecutive-prime bias. The mechanism is the bias of Lemke Oliver and Soundararajan in consecutive prime residues, propagated through the factorisation of $p - 1$ to the Artin property; as an intermediate link we measure the (apparently unreported) anticorrelation $r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big) = -0.041$. We formulate quantitative conjectures and discuss extensions to general bases.

1. Introduction

Two classical themes in the theory of primes intersect in this paper.

The first is Artin's primitive root conjecture (1927): for any integer $a \notin \{-1, 0, 1\}$ that is not a perfect square, the set of primes p for which a is a primitive root modulo p has positive density; for base $a = 10$ this density is Artin's constant $C = \prod_q \big(1 - \frac{1}{q(q-1)}\big) \approx 0.3739558$. Hooley [11] proved the conjecture under the Generalized Riemann Hypothesis, Gupta and Murty [8] and Heath-Brown [10] obtained the strongest unconditional results, and Moree's survey [14] gives a comprehensive account. For $a = 10$ the property has a pleasant classical meaning: 10 is a primitive root mod p precisely when the decimal expansion of $1/p$ has maximal period $p - 1$ (the *full reptend* or *long* primes; see [3, pp. 166–171]).

The second theme is the surprising dependence between *consecutive* primes discovered by Lemke Oliver and Soundararajan [12]: for a modulus q, a prime in a given reduced residue class mod q is followed by a prime in the *same* class markedly less often than equidistribution would predict. The phenomenon, which they explain conjecturally via the Hardy–Littlewood prime k-tuple conjectures [9], decays slowly and is prominent at the scales accessible to computation.

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This paper asks a question that appears not to have been raised before: *are the Artin statuses of consecutive primes correlated?* Since the Artin property of p is governed by the arithmetic of $p-1$ (which primes q divide it), and since the residues of consecutive primes modulo small q are correlated by [12], one expects *some* induced dependence between $\mathbf{1}[10$ is a primitive root mod $p_n]$ and $\mathbf{1}[10$ is a primitive root mod $p_{n+1}]$. What is not obvious is its sign, its size, its structure as a function of the gap $g = p_{n+1} - p_n$, or how completely it is accounted for by known mechanisms. We answer these questions computationally for all primes up to 10^9 and isolate one component that is not merely statistical but *deterministic*: an exclusion law for gaps $g \equiv 20 \pmod{40}$ (Theorem 1).

Related but distinct questions have been studied. Garcia, Kahoro, and Luca [6] (see also the sequel [7]) compared the *number* of primitive roots, $\varphi(p-1)$ versus $\varphi(p+1)$, for twin prime pairs, finding a strong bias which they explain under the Bateman–Horn conjecture. Pollack [16] and Baker and Pollack [2] proved, by combining Hooley's method with the Maynard–Tao sieve [13, 17], that (under GRH) there are infinitely many runs of m *consecutive* primes all having a prescribed primitive root within a bounded interval. We emphasise that these are *existence* results, produced by sieve constructions that select rare favourable configurations; they make no statement about the typical joint behaviour of Artin status along the sequence of primes. The present paper asks the complementary *statistical* question — how the Artin statuses of consecutive primes are correlated on average — which appears to be unexamined; and the exclusion law of Theorem 1, though elementary, seems to be unrecorded.

Recent work has also sharpened the analytic understanding of Artin densities for *individual* primes: Fan and Pollack [5] establish (under GRH) a version of the Artin–Hooley asymptotic that is uniform in the base, and Perucca and Shparlinski [15] prove uniform upper and lower bounds for the Artin density relative to the corresponding Artin constant. These results concern the distribution of Artin primes one prime at a time; to our knowledge the *joint* behaviour of the Artin property at consecutive primes, which is the subject of this paper, has not been considered.

Summary of results.

(1) (Theorem 1) If p and $p+g$ are primes exceeding 5 and $g \equiv 20 \pmod{40}$, then $\left(\frac{10}{p+g}\right) = -\left(\frac{10}{p}\right)$; consequently 10 is a quadratic residue for exactly one of the two, and at most one can be an Artin prime for base 10. If instead $g \equiv 0 \pmod{40}$, then $\left(\frac{10}{p+g}\right) = \left(\frac{10}{p}\right)$.

(2) (Section 4) Across all 50,847,529 consecutive prime pairs with $7 \le p \le 10^9$, Artin statuses are anticorrelated: $\delta = P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) - P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n) = -0.01414$, with $\chi^2 =$ 10,167 on one degree of freedom.

(3) (Section 5) The dependence is violently non-uniform in the gap: $\delta(g)$ ranges from -0.553 (at $g = 20$, where the theorem forces exclusion) to $+0.643$ (at $g = 40$), with sign and magnitude predicted by $g \bmod 40$ and $g \bmod 3$.

(4) (Section 6) Conditioning on the pair of residues of (p_n, p_{n+1}) modulo 120 shrinks the mean dependence by a factor of 13; modulo 840, by a factor of 30, to $\delta_{\mathrm{res}} = -0.00047$ with $\chi^2/\mathrm{df} = 1.14$ — consistent with the residue channels at moduli beyond 840 and with no additional mechanism.

(5) (Section 7) As an intermediate link in the causal chain we measure $r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big) = -0.0410$: consecutive primes' $p-1$ factorisations repel.

Throughout, *Artin prime* means Artin prime for base 10 unless stated otherwise, and p_n denotes the n-th prime.

2. The exclusion law

Definition 1. For an odd prime $p \nmid 10$, write $\left(\frac{10}{p}\right)$ for the Legendre symbol. We say p is an *Artin prime* (for base 10) if the multiplicative order of 10 modulo p equals $p-1$.

If 10 is a quadratic residue mod p then its order divides $(p-1)/2$, so a QR is never a primitive root; $\left(\frac{10}{p}\right) = -1$ is a necessary (not sufficient) condition for the Artin property.

Theorem 1 (Exclusion law for gaps $\equiv 20 \pmod{40}$)**.** *Let p and $p' = p + g$ be primes with $p > 5$ and $g \equiv 20 \pmod{40}$. Then*

$$\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right).$$

In particular, 10 is a quadratic residue modulo exactly one of p, p', and at most one of them is an Artin prime for base 10.

Proof. By multiplicativity, $\left(\frac{10}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{5}{p}\right)$.

Since $g \equiv 0 \pmod 5$, we have $p' \equiv p \pmod 5$. By quadratic reciprocity ($5 \equiv 1$ mod 4), $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$ depends only on p mod 5; hence $\left(\frac{5}{p'}\right) = \left(\frac{5}{p}\right)$.

Since $g \equiv 4 \pmod 8$, we have $p' \equiv p + 4 \pmod 8$. The map $r \mapsto r + 4$ on odd residues mod 8 exchanges $1 \leftrightarrow 5$ and $3 \leftrightarrow 7$. By the second supplementary law, $\left(\frac{2}{p}\right) = +1$ iff $p \equiv \pm 1 \pmod 8$, and the exchange maps $\{1, 7\}$ onto $\{5, 3\}$ and vice versa; hence $\left(\frac{2}{p'}\right) = -\left(\frac{2}{p}\right)$ for every admissible residue.

Multiplying, $\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right)$. The prime with $\left(\frac{10}{\cdot}\right) = +1$ has $\operatorname{ord}(10) \mid (p-1)/2 < p - 1$, so it is not an Artin prime. □

Corollary 1 (Coupling law for gaps $\equiv 0 \pmod{40}$)**.** *If $p, p' = p+g$ are primes with $p > 5$ and $g \equiv 0 \pmod{40}$, then $\left(\frac{10}{p'}\right) = \left(\frac{10}{p}\right)$. In particular, if either is an Artin prime, then 10 is a quadratic non-residue for both.*

Proof. Now $g \equiv 0 \pmod 8$ and $g \equiv 0 \pmod 5$, so both $\left(\frac{2}{\cdot}\right)$ and $\left(\frac{5}{\cdot}\right)$ are preserved. □

Remark 1. *The modulus 40 is specific to base 10: it is the conductor of the quadratic character $p \mapsto \left(\frac{10}{p}\right)$, namely $4 \cdot 10 = 40$ (the discriminant of $\mathbb{Q}(\sqrt{10})$). For a general non-square base a the same argument applies with 40 replaced by the conductor $|d_a|$ of the quadratic character attached to $\mathbb{Q}(\sqrt{a})$: gaps $g \equiv 0 \pmod{|d_a|}$ preserve quadratic-residue status, while suitable classes flip it. Base 10 is distinguished only by our decimal habit and the full-reptend interpretation.*

Remark 2. *For gaps in classes other than $0, 20 \pmod{40}$ the relation between $\left(\frac{10}{p}\right)$ and $\left(\frac{10}{p'}\right)$ depends on p as well as g (the shift on p mod 8 or p mod 5 is not a symbol-preserving or symbol-flipping involution), producing the partial correlations catalogued in Section 5.*

We verified Theorem 1 computationally as a guard against error: for all 23,956 consecutive prime pairs below 10^7 with gap $\equiv 20 \pmod{40}$, the Legendre symbols flip in every single case, and a direct order computation on a systematic subsample found no doubly-Artin pair. In the full dataset up to 10^9 (Section 5), the 2,195,882 consecutive pairs with $g \in \{20, 60\}$ contain *no* pair in which both primes are Artin — an empirical count of zero, exactly as the theorem requires.

3. Data and computation

We use the dataset of our companion computation: for every prime $7 \le p \le 10^9$ (50,847,531 primes, generated by a segmented sieve of Eratosthenes in C), we store the gap to the neighbouring primes, the number $\omega(p-1)$ of distinct prime factors of $p-1$ (by trial division), the Artin indicator (computed exactly via $10^{(p-1)/q} \not\equiv 1 \pmod p$ for each prime $q \mid p-1$, using GMP modular exponentiation), and small residues of p. As a global check, the number of Artin primes in the dataset is 19,016,618, a proportion 0.373993, matching Artin's constant 0.373956 to four decimal places (for base 10 the squarefree part is $10 \not\equiv 1 \bmod 4$, so no correction factor applies and the conjectural density is exactly C; see [11, 14]).

The consecutive-pair analysis is a single streaming pass over the 50,847,529 ordered pairs (p_n, p_{n+1}), accumulating 2×2 contingency tables of $(\mathbf{1}[\mathrm{Art}_n], \mathbf{1}[\mathrm{Art}_{n+1}])$: globally, stratified by gap g, and stratified by joint residues of (p_n, p_{n+1}) to various moduli. All counts reported below are exact (no sampling), except where noted. Analysis scripts (Python) and their outputs are available with the data (see the Data Availability statement).

4. The global anticorrelation

Table 1. Artin status of consecutive primes, all 50,847,529 pairs with $7 \le p_n \le 10^9$.

Quantity	Value
$P(\mathrm{Art}_{n+1})$	0.373993
$P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n)$	0.365141
$P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n)$	0.379281
$\delta =$ difference	-0.014140
ϕ coefficient	-0.014140
χ^2 (1 df)	10,166.7
z-score of δ	-100.8

Table 1 presents the central fact: an Artin prime is followed by another Artin prime about 1.4 percentage points less often than a non-Artin prime is. The effect is small in absolute terms but enormous relative to sampling noise. If Artin statuses of consecutive primes were independent, the observed $|\delta|$ would be a $\sim 100\sigma$ event.

The sign is what the Lemke Oliver–Soundararajan repulsion predicts. Heuristically: the Artin property requires $\left(\frac{10}{p}\right) = -1$, which is a condition on $p \bmod 40$; it is further favoured by $3 \nmid p-1$, i.e. $p \equiv 2 \pmod 3$, since a factor of 3 in $p-1$ contributes the largest single suppression $(1 - \frac{1}{3})$ of the primitive-root density among odd primes. Consecutive primes repel in their residues modulo 40 and modulo 3 [12]; conditions favourable to the Artin property therefore tend *not* to repeat at p_{n+1} when they hold at p_n, depressing $P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n)$.

For reference, the mod-12 transition matrix in our data displays the repulsion prominently: the diagonal entries $P(p_{n+1} \equiv r \mid p_n \equiv r)$ are 0.1813–0.1815 for $r \in \{1, 5, 7, 11\}$, against the equidistribution value 0.25 — a 27% deficit, consistent in magnitude with the computations in [12] at comparable scales.

5. Gap structure

Stratifying by the gap $g = p_{n+1} - p_n$ reveals that the global $\delta = -0.0141$ is the average of far larger, sign-alternating effects. Table 2 shows all gaps with at least 10^5 pairs; Figure 1 displays $\delta(g)$ with the residue class of $g \bmod 40$ marked.

Three features stand out.

(i) The exclusion rows. At $g = 20$ (1,824,043 pairs) and $g = 60$ (371,839 pairs), $P(A \mid A)$ is exactly zero: among 2.2 million consecutive pairs, not one is doubly Artin. This is Theorem 1 in action, and conversely the observation that motivated its proof.

(ii) The $g = 40$ spike. $P(A \mid A) = 0.899$, the largest conditional probability in the table and 2.4 times the unconditional density. Two forced mechanisms combine. First, by Corollary 1, if p_n is Artin then 10 is a non-residue for p_{n+1} as well, doubling its prior ($1/C$·-fold, roughly, since about half of primes have $\left(\frac{10}{p}\right) = -1$). Second, $40 \equiv 1 \pmod 3$ forces the pair's mod-3 residues: $p_n \equiv 2$

TABLE 2. Conditional Artin dependence by gap ($N \geq 10^5$ pairs). $\delta(g) = P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) - P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n)$.

g	N	$P(A\|A)$	$P(A\|\neg A)$	$\delta(g)$	g	N	$P(A\|A)$	$P(A\|\neg A)$	$\delta(g)$
2	3,424,504	0.291	0.293	−0.002	32	683,896	0.176	0.386	−0.209
4	3,424,679	0.562	0.381	+0.181	34	719,531	0.445	0.445	+0.001
6	6,089,791	0.407	0.382	+0.025	36	1,171,524	0.559	0.314	+0.245
8	2,695,108	0.160	0.409	−0.249	38	548,746	0.288	0.293	−0.006
10	3,484,767	0.448	0.443	+0.005	40	648,780	0.899	0.257	+0.643
12	4,468,957	0.512	0.283	+0.229	42	954,456	0.380	0.358	+0.023
14	2,464,565	0.298	0.296	+0.002	44	389,634	0.344	0.249	+0.095
16	1,846,097	0.296	0.543	−0.247	46	334,720	0.469	0.469	+0.000
18	3,351,032	0.378	0.359	+0.019	48	577,247	0.216	0.454	−0.238
20	1,824,043	**0.000**	0.553	−0.553	50	328,066	0.312	0.307	+0.005
22	1,569,679	0.444	0.447	−0.004	52	245,804	0.572	0.381	+0.191
24	2,367,474	0.295	0.411	−0.116	54	410,754	0.382	0.356	+0.026
26	1,119,585	0.313	0.313	+0.000	56	211,462	0.225	0.389	−0.164
28	1,219,243	0.619	0.366	+0.253	58	181,948	0.437	0.441	−0.004
30	2,177,991	0.390	0.363	+0.027	60	371,839	**0.000**	0.592	−0.592

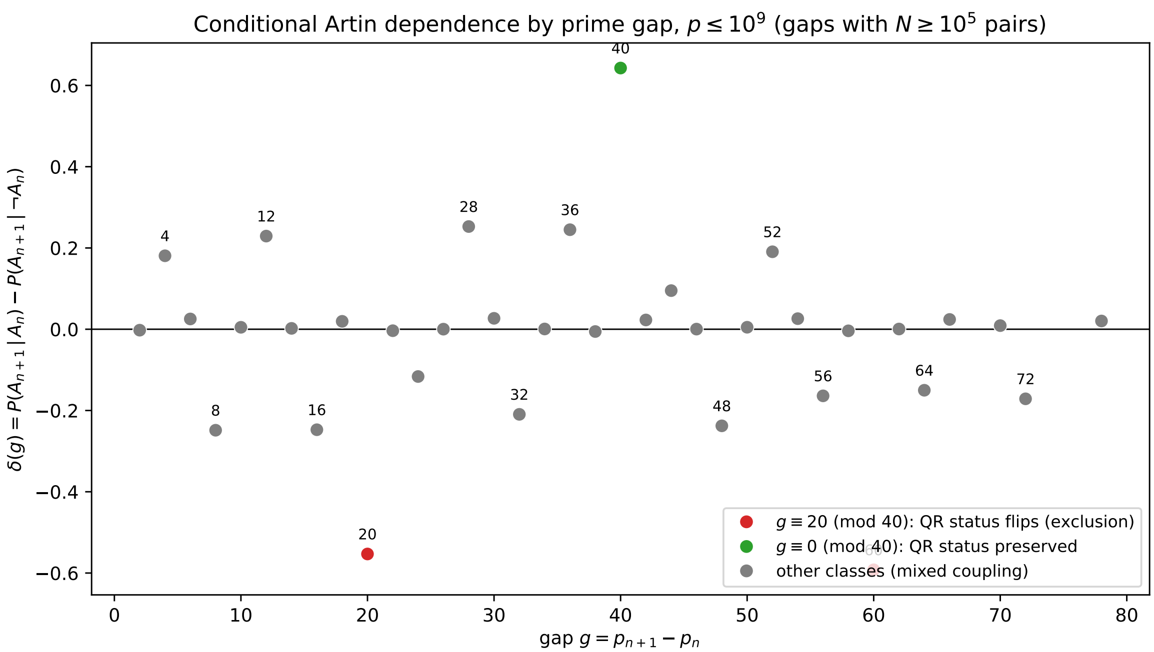

FIGURE 1. $\delta(g)$ for all gaps with $N \geq 10^5$ pairs, $p \leq 10^9$. Red: $g \equiv 20 \pmod{40}$ (Theorem 1 forces $P(A|A) = 0$). Green: $g \equiv 0 \pmod{40}$ (Corollary 1 preserves quadratic-residue status). Grey: other classes.

(mod 3) would give $3 \mid p_n + 40$, impossible; so every $g = 40$ pair has $p_n \equiv 1$, $p_{n+1} \equiv 2 \pmod 3$ — that is, $3 \mid p_n - 1$ (suppressing Art_n) and $3 \nmid p_{n+1} - 1$ (favouring Art_{n+1}). Conditioning on Art_n therefore selects pairs in which p_{n+1} enjoys both the non-residue status and the mod-3 advantage, yielding the observed 0.899.

(iii) Sign classes. Gaps divisible by 3 but not constrained mod 40 (e.g. $6, 18, 30, 42, 54$) show small positive $\delta \approx +0.02$: the pair shares its mod-3 status, weakly coupling the largest single suppression factor. Gaps $g \equiv 8, 16 \pmod{24}$-type classes with adverse mod-8/mod-5 shifts show strong negative δ. The entire sign pattern of Table 2 is reproduced by tracking how the shift by g permutes the residues of p modulo 8, 5, and 3; no free parameter is involved.

6. Decomposition of the global effect

How much of the global $\delta = -0.0141$ do the explicit residue channels account for? We condition on the *joint* residue class of (p_n, p_{n+1}) to modulus m and measure the residual within-cell dependence, aggregating cells by Cochran–Mantel–Haenszel-style summation of per-cell χ^2 (that is, testing the within-cell association jointly across all cells, so that dependence explained by the conditioning variable itself contributes nothing) and by the pair-weighted mean of per-cell δ.

Conditioning on $m = 120 = \mathrm{lcm}(8, 3, 5)$ fixes the quadratic character $\left(\frac{10}{\cdot}\right)$ of both primes (its conductor 40 divides 120) and the mod-3 channel. Cells in which 10 is a quadratic residue for p_n or p_{n+1} are degenerate for the analysis (one margin of the 2×2 table vanishes — the Artin property is impossible); the informative population is the quarter of pairs in doubly-non-residue cells.

Table 3. Residual dependence after conditioning on joint residues mod m. "δ_{res}" is the pair-weighted mean within-cell δ; cells with fewer than 5000 ($m = 120$) or 2000 ($m = 840$) pairs are excluded from the summed χ^2.

Conditioning	cells	pairs used	$\sum \chi^2$ (df)	δ_{res}
none (global)	1	50,847,529	10,166.7 (1)	−0.014140
mod 120 joint residues	145	12,255,203	789.2 (145)	−0.001055
mod 840 joint residues	633	12,086,765	719.8 (633)	−0.000473

Conditioning mod 120 removes 93% of the mean dependence; extending to $840 = \mathrm{lcm}(8, 3, 5, 7)$, which additionally fixes whether $7 \mid p - 1$ on both sides, removes 97% and brings the aggregate χ^2 per degree of freedom down to $719.8/633 = 1.14$, close to the null value 1. Two consistency checks support the interpretation that the remaining crumb of dependence is simply the tail of the same mechanism at moduli beyond 840 (channels $q = 11, 13, \ldots$), rather than anything new: (i) the residual shrinks monotonically, and roughly geometrically, as channels are absorbed ($-0.0141 \to -0.00106 \to -0.00047$); (ii) a split-sample check (pairs with $p < 5 \times 10^8$ versus $p \geq 5 \times 10^8$, conditioning mod 120) gives residuals -0.00096 and -0.00094 — stable across the range, as a residue-driven effect should be, whereas a finite-size artefact would be expected to shrink.

We summarise the decomposition as follows: *the consecutive-prime Artin anticorrelation is, to the accuracy measurable at* 10^9*, entirely a sum of residue-coupling channels: the quadratic-character channel at conductor* 40*, and one channel for each small prime* q*, each transmitting the Lemke Oliver–Soundararajan bias at modulus* q *into the divisibility of* $p-1$ *by* q*, and hence into the Artin property.*

7. The intermediate link: repulsion of $\omega(p-1)$

The chain from residue bias to Artin status passes through the factorisation of $p - 1$. Its first summary statistic is $\omega(p-1)$, the number of distinct prime factors, whose average grows like $\log \log p$ (for $p - 1$ shifted primes this is a classical result of Erdős [4]). If consecutive primes repel modulo every small q, then $q \mid p_n - 1$ makes $q \mid p_{n+1} - 1$ *less* likely than average, and summing over q one predicts a negative correlation between $\omega(p_n - 1)$ and $\omega(p_{n+1} - 1)$.

The data confirm this with a value that is, to our knowledge, unreported:

$$r\big(\omega(p_n - 1),\, \omega(p_{n+1} - 1)\big) = -0.0410 \qquad (50{,}847{,}529 \text{ pairs},\ p \leq 10^9).$$

The factorisations of $p-1$ for consecutive primes are not independent: they repel, in the same sense and for the same reason that the primes themselves repel in residue classes.

8. Conjectures

The Hardy–Littlewood-based framework of [12] makes the residue transition frequencies of consecutive primes quantitatively predictable, and every channel in our decomposition is built from such frequencies. This motivates:

Conjecture 1 (Persistence and decay)**.** *For primes up to x, the consecutive-pair Artin correlation $\delta(x) = P(\mathrm{Art}_{n+1} \mid \mathrm{Art}_n) - P(\mathrm{Art}_{n+1} \mid \neg\mathrm{Art}_n)$ is negative for all sufficiently large x, tends to 0 as $x \to \infty$, and does so at the (slow) rate of the Lemke Oliver–Soundararajan bias, i.e. $\delta(x) \asymp -1/\log x$ up to factors of $\log\log x$.*

The heuristic for the rate: each channel's transition bias decays like $(\log x)^{-1}$ times slowly varying corrections [12], and the channels combine linearly at first order. At $x = 10^9$ we measure $\delta = -0.0141$; the conjecture predicts, for instance, $|\delta| \approx 0.010$–0.012 at $x = 10^{12}$, a directly testable statement.

Conjecture 2 (Completeness of the residue decomposition)**.** *For every $\varepsilon > 0$ there is a modulus M such that conditioning on the joint residues of (p_n, p_{n+1}) modulo M reduces the weighted mean within-cell dependence below ε in absolute value, uniformly in x. Equivalently: consecutive-prime Artin correlation is carried entirely by joint residue distributions, with no additional mechanism.*

Conjecture 3 (ω repulsion)**.** *$r\big(\omega(p_n - 1), \omega(p_{n+1} - 1)\big)$ is negative for all large x and tends to 0 as $x \to \infty$.*

We remark that Conjecture 2 is in the spirit of — and would follow from a sufficiently uniform version of — the Hardy–Littlewood prime pair conjectures, since the joint distribution of $(p_n \bmod M, p_{n+1} \bmod M, g)$ is governed by singular series; but we state it as an empirical proposition because that is what the data test.

9. Discussion

What is new here. The individual ingredients are classical: quadratic reciprocity, the QR-obstruction to primitive roots, and the consecutive-prime bias of [12]. The contributions of this paper are (i) the observation that these combine into a deterministic exclusion law for prime pairs at gaps $\equiv 20 \pmod{40}$ (Theorem 1) — elementary, but we have not found it stated anywhere; (ii) the first measurement of the consecutive-prime Artin correlation, its gap profile, and the ω repulsion; and (iii) the demonstration that the measured dependence decomposes, quantitatively and apparently completely, into residue-coupling channels. Point (iii) is the analogue, for a multiplicative property of $p-1$, of the explanation of the original bias via the Hardy–Littlewood conjectures, and turns the Artin correlation from a curiosity into a sharp consistency test of that framework at scale.

Relation to runs of Artin primes. Pollack [16] and Baker–Pollack [2] show (under GRH) that infinitely many runs of m consecutive primes share a primitive root lying in a bounded interval. Our results add a quantitative caution to the base-specific version of such statements: for base 10, runs of consecutive Artin primes must navigate the exclusion law — a run is broken whenever a gap $\equiv 20 \pmod{40}$ occurs — and the prevailing anticorrelation makes long runs rarer than independence would suggest. It would be interesting to quantify the interaction between the sieve constructions and the residue channels identified here.

Future work. Three extensions suggest themselves. (1) Other bases: the same pipeline with $a = 2, 3, 5, 6, 7$ would test the conductor prediction of the Remark after Corollary 1 (e.g. conductor 8 for $a = 2$, so exclusion classes mod 8; conductor 24 for $a = 6$). (2) Scale: extending to 10^{12} tests the decay rate of Conjecture 1. (3) Theory: a derivation of $\delta(g)$ from the singular series of [9] combined with the density computations of Hooley's method, at least for the dominant channels, appears feasible and would replace our empirical decomposition with a formula.

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Data availability

All code required to reproduce every number in this paper is publicly available at `https://github.com/jpbald93/consecutive-artin` (archived at the time of submission). The repository contains the C segmented sieve that generates the per-prime dataset, the Python analysis scripts, the figure script, and the raw output logs. The per-prime dataset itself (1.8 GB CSV covering all primes to 10^9: gap, $\omega(p-1)$, Artin indicator, and small residues for each prime) is not deposited because of its size, but it is exactly and deterministically reproducible from the included C source in a few hours on a single machine; it is also available from the author on request.

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